

Math 147

Analysis

Qinghao Hu

September 9, 2026

Contents

1 Course Notes 3

1.1 General Information 3

1.1.1 Contact 3

1.1.2 Course Structure 3

2 Introduction to Proof 4

2.1 Lecture 1 4

2.1.1 What Is Calculus? 4

2.1.2 Numbers 4

1 Course Notes

1.1 General Information

1.1.1 Contact

Email: mmolino@uwaterloo.ca

Website: www.math.uwaterloo.ca/~mmolino

Office hours: Monday, 3:30–4:30 p.m., and Friday, 10:00–11:00 a.m., in MC 4006.

1.1.2 Course Structure

Lectures: Section 002 meets on Monday, Wednesday, and Friday from 12:30–1:20 p.m. in MC 4021.

2 Introduction to Proof

2.1 Lecture 1

2.1.1 What Is Calculus?

Differential and integral calculus are two branches of calculus.

Differential calculus: The study of slopes and tangent lines.

Integral calculus: The study of areas and accumulation.

2.1.2 Numbers

The number systems discussed in this lecture include

$$\begin{aligned}\mathbb{N} &= \{1, 2, 3, \dots\}, \\ \mathbb{Z} &= \{\dots, -1, 0, 1, \dots\}, \\ \mathbb{Q} &= \left\{ \frac{m}{n} : m, n \in \mathbb{Z}, n \neq 0 \right\}.\end{aligned}$$

The natural numbers are closed under addition. The integers are closed under addition, subtraction, and multiplication, but they are not closed under division; for example, $1/2 \notin \mathbb{Z}$. Therefore, we introduce the rational numbers to include quotients of integers with nonzero denominators.

Theorem 2.1 (Irrationality of $\sqrt{2}$). *The number $\sqrt{2}$ is irrational; equivalently, $\sqrt{2} \notin \mathbb{Q}$.*

Proof. Suppose, for contradiction, that $\sqrt{2} \in \mathbb{Q}$. Then we can write

$$\sqrt{2} = \frac{m}{n}, \quad m, n \in \mathbb{Z}, \quad n > 0, \quad \gcd(m, n) = 1.$$

Squaring both sides gives

$$2 = \frac{m^2}{n^2} \quad \implies \quad m^2 = 2n^2.$$

Thus, m^2 is even, so m is even. Write $m = 2\ell$ for some $\ell \in \mathbb{Z}$. Substituting this into the equation above gives

$$\begin{aligned}2n^2 &= 4\ell^2, \\ n^2 &= 2\ell^2.\end{aligned}$$

Therefore, n^2 is even, and hence n is even as well.

This means that both m and n are even, contradicting $\gcd(m, n) = 1$. Therefore, $\sqrt{2} \notin \mathbb{Q}$. ■

The irrational numbers are the real numbers that are not rational:

$$\mathbb{R} \setminus \mathbb{Q} = \{x \in \mathbb{R} : x \notin \mathbb{Q}\}.$$

Example 2.2 (Decimal representation). Every real number has a decimal representation. In general, a decimal expansion can be written as

$$x = a + \sum_{i=1}^{\infty} \frac{x_i}{10^i}, \quad a \in \mathbb{Z}, \quad x_i \in \{0, 1, 2, \dots, 9\}.$$